

COM 3240

REINFORCEMENT LEARNING

MINIMAL MATHEMATICAL PREREQUISITES

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Wikipedia - Plato's Academy mosaic



Villa of T. Siminius Stephanus, Pompeii

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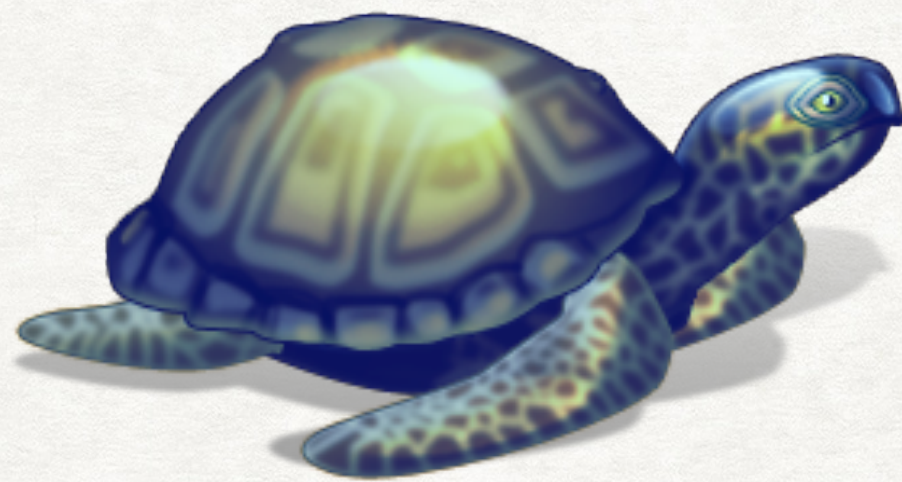
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Αεί ὁ Θεός ὁ Μέγας γεωμετρεῖ.

— *Attributed to Plato*

”

Great God always geometrizes



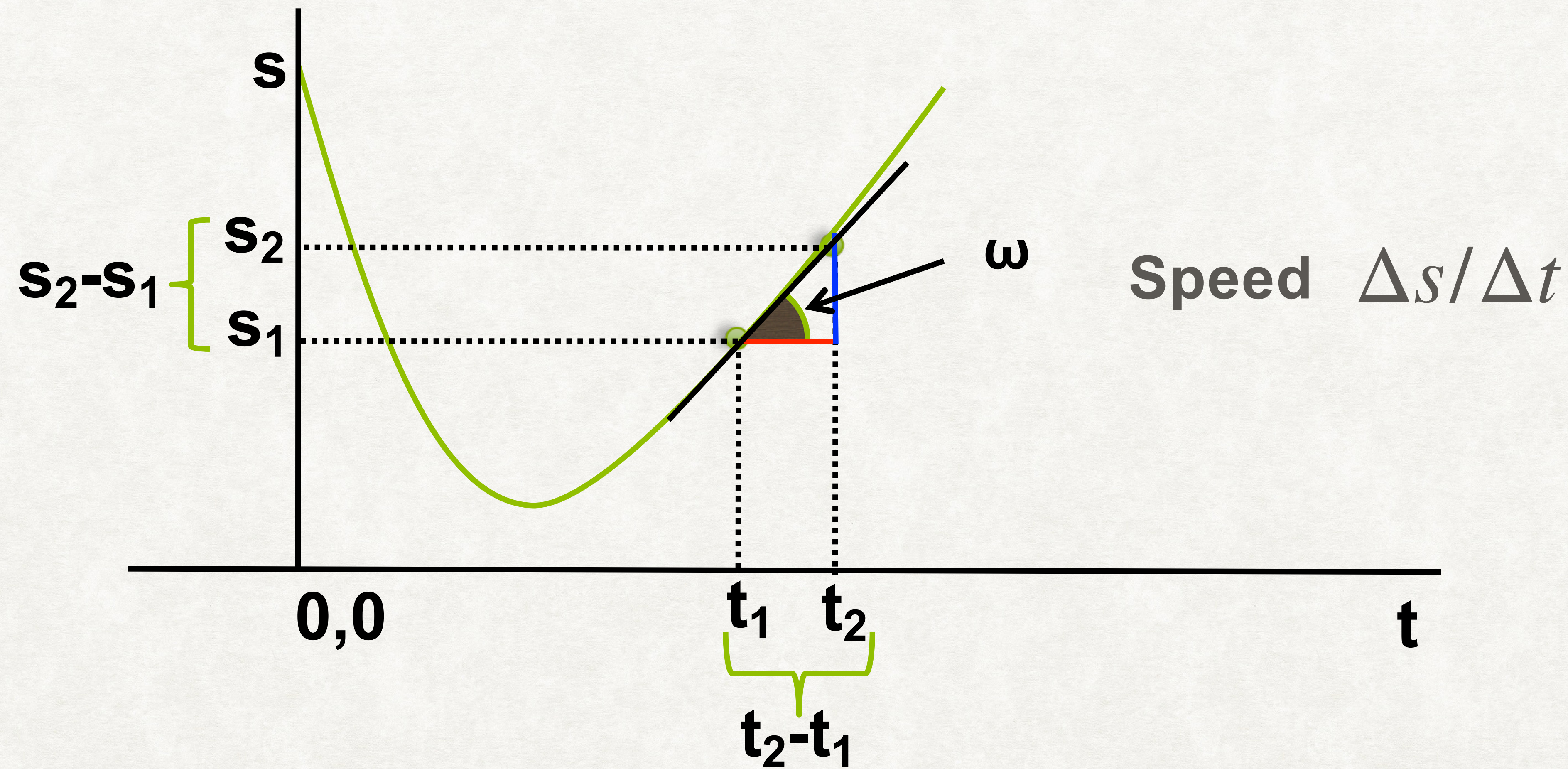
s_1 **s_2**

Speed $\Delta s / \Delta t$

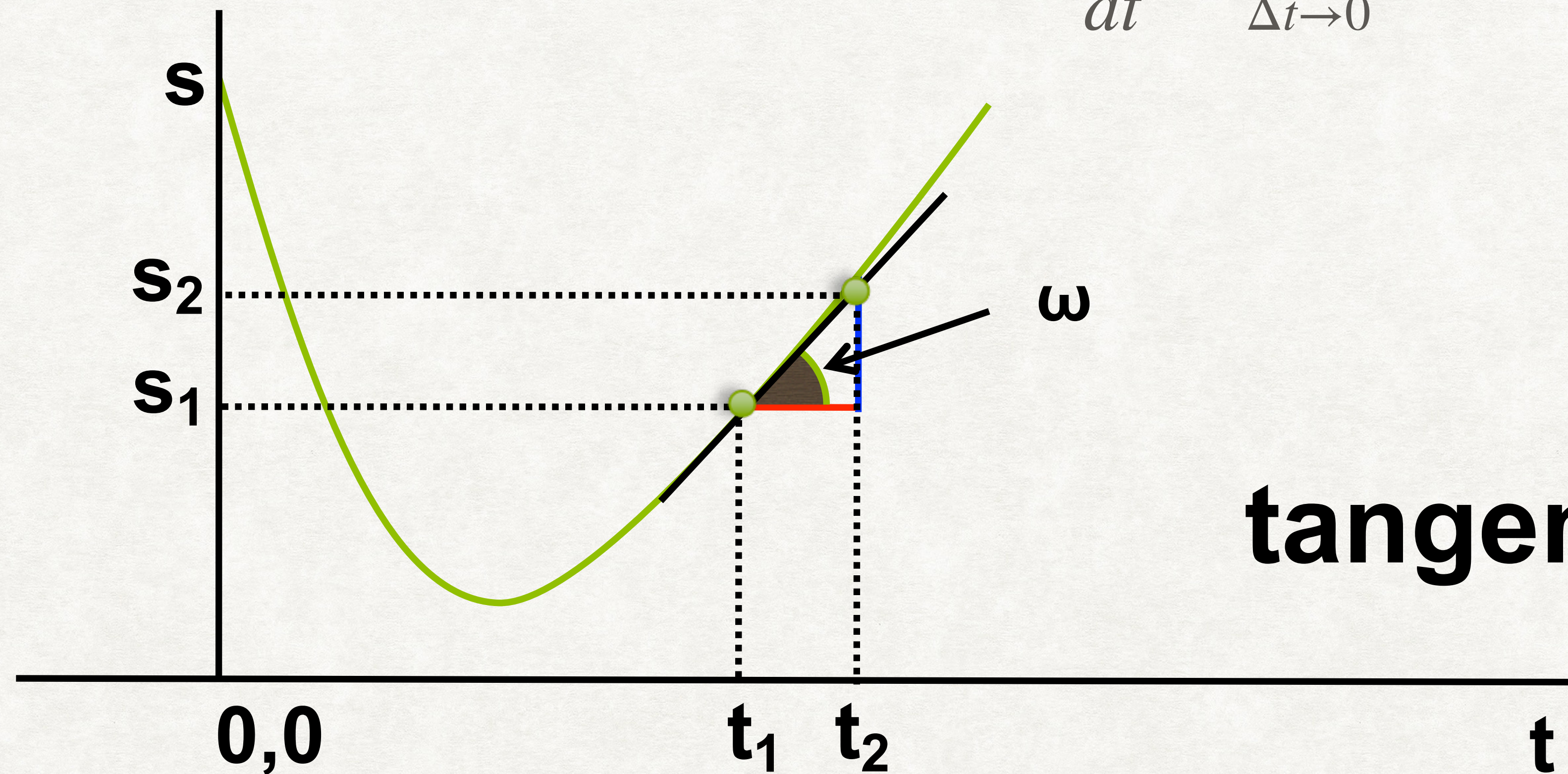
$$\Delta s = s_2 - s_1$$

t_1 **t_2**

$$\Delta t = t_2 - t_1$$



$$\frac{ds}{dt} = \lim_{\Delta t \rightarrow 0} \frac{s(t + \Delta t) - s(t)}{\Delta t}$$



tangent ω

DERIVATIVES

USE THE DEFINITION TO CALCULATE DERIVATIVES OF FUNCTIONS

More generally:

$$f'(x) = \frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

What is the effect of incrementing x by a small amount, Δx , on the value of $f(x)$?

$$f(x) = 7$$

$$f(x) = 7x$$

$$f(x) = 7x^2$$

DERIVATIVES IN PRACTICE

MEMORISE BASIC FUNCTIONS AND USE RULES

Power function: $f(x) = x^n$ $f'(x) = nx^{n-1}$

Exponential function: $f(x) = e^x$ $f'(x) = e^x$

Natural logarithm: $f(x) = \ln(x)$ $f'(x) = \frac{1}{x}$

DERIVATIVES - RULES

ADDITION

$$\frac{d}{dx} (f(x) + g(x)) = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

$$f(x) = x + x^2$$

DERIVATIVES - RULES

MULTIPLICATION

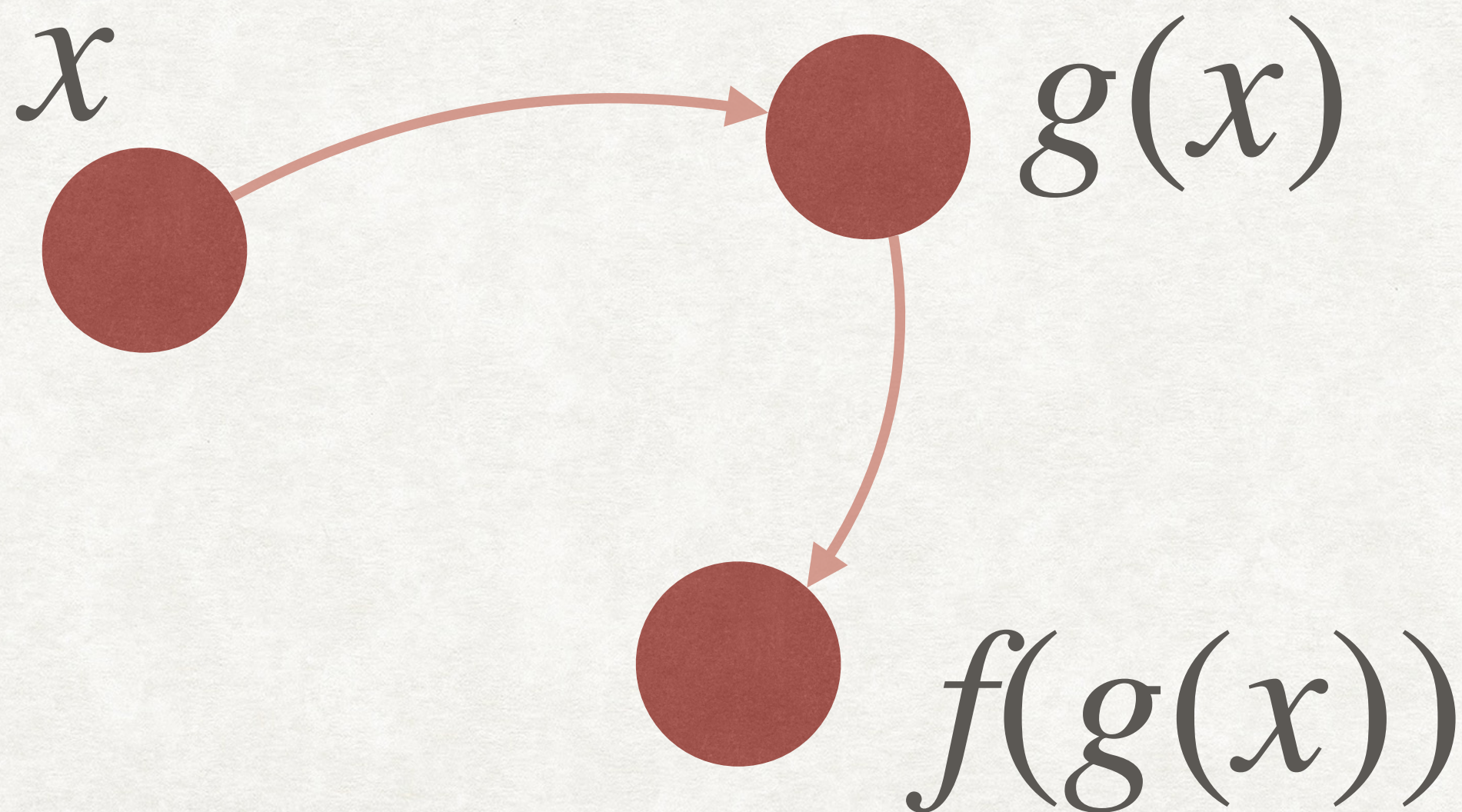
$$\frac{d}{dx} (f(x) \cdot g(x)) = \frac{df(x)}{dx} \cdot g(x) + f(x) \cdot \frac{dg(x)}{dx}$$

$$f(x) = x \cdot \ln(x)$$

DERIVATIVES - RULES

THE CHAIN RULE

$$\frac{df(g(x))}{dx} = \frac{df(g(x))}{dg(x)} \cdot \frac{dg(x)}{dx}$$



$$f(x) = (x - c)^2$$

PARTIAL DERIVATIVES

AND THE NABLA NOTATION

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y, \dots) - f(x, y, \dots)}{\Delta x}$$

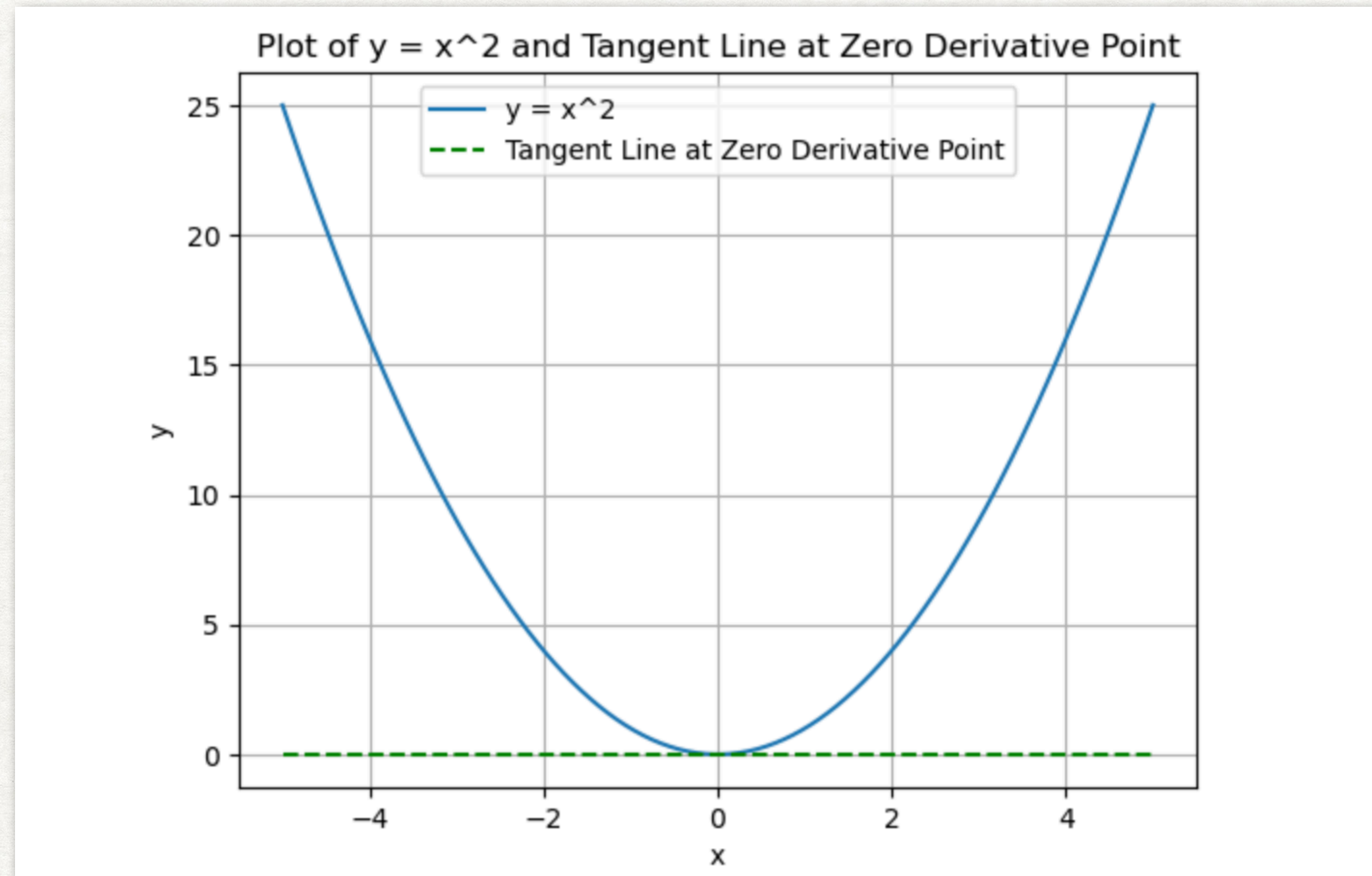
$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2y + y^2) = 2xy$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^2y + y^2) = x^2 + 2y$$

DERIVATIVES

OPTIMISATION - GRADIENT METHODS



Error (Loss) function - Machine Learning

DEFINITION OF DERIVATIVES

AND ITS APPLICATION TO DIFFERENTIAL EQUATIONS

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\tau \frac{d\bar{x}}{dt} = -\bar{x}(t) + s(t) \quad \text{s: sample}$$

Its discrete form is known as “exponential moving average”

THANK YOU!